

may result in a smaller (or larger) drop. Swing traders may drool at capturing such a decline, but buy-and-hold investors may yawn.

Standard & Poor's 500 change. Tracking the performance of the S&P using the same hold time from breakout to ultimate low as the pipes shows that the market declined along the way. The market decline helped the individual stocks drop.

Days to ultimate low. The average decline in bear markets is 24% and takes 39 days. In bull markets, the decline is 19% and takes 53 days. Thus, the bear market decline is faster than the bull market one. Doing the math, we find the bear market decline is 1.7 times faster than the bull market drop.

How many change trend? This is a count of how many pipes see price drop more than 20%. Values for *bullish* (not pipe tops) patterns over 50% I consider terrific, but I don't have a gauge for bearish patterns. Clearly the bear market number is wonderful, and the bull market rate isn't bad, either.

[Table 50.3](#) shows failure rates for pipes (this sounds like something to do with plumbing). Notice that the failure rates in bear markets are lower than are those in bull markets. For a bearish pattern, that's no surprise.

Let me provide a few examples so you understand how to read the table. In bear markets, 5% of the pipes will fail to see price drop more than 5%. That rate more than triples in the next row when we see 19% fail to see price drop more than 10%. Almost half of the patterns (48%) won't exceed a 20% drop.

Notice how the failure rates climb for declines of 5%, 10%, and 15%. The rapid increase in failures for small price changes is typical for chart patterns. It emphasizes the risky nature of chart pattern trading and the difficulty of making large gains.

[Table 50.4](#) shows breakout-related statistics.